

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/2

PAPER 2 November 2018

2 hours

PAPER 2 — STRUCTURED QUESTIONS — ANSWER ALL — 2 HOURS

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2018 5006/2.

Additional materials: Answer all questions. The number of marks is given in brackets [].

INSTRUCTIONS: Write your answers in the spaces provided. You may use a calculator.

1. (a) Define diffusion. [2]<<
>>(b) Define osmosis. [2]<<
>>(c) A peeled potato chip is placed in concentrated sugar solution. Describe and explain what happens to its mass. [3]<<
>>(d) Explain why root hair cells have a large surface area. [3] **[10]**

2. (a) Write the balanced symbol equation for photosynthesis. [3]<<
>>(b) Name the green pigment and the organelle where it is found. [2]<<
>>(c) State two factors that can limit the rate of photosynthesis. [2]<<
>>(d) Describe a test for starch in a destarched leaf that has been exposed to light. [3] **[10]**

3. (a) What is a pathogen? [2]<<
>>(b) Distinguish between an antigen and an antibody. [2]<<
>>(c) Explain how a vaccine can protect a person against a disease. [4]<<
>>(d) Suggest why antibiotics are not used to treat a viral infection such as influenza. [2] **[10]**

4. (a) Name the type of pathogen that causes cholera. [1]<<
>>(b) Describe how cholera is transmitted. [2]<<
>>(c) Explain why oral rehydration is used to treat cholera. [3]<<
>>(d) HIV is a virus. Explain how it leads to AIDS, and state one way transmission can be reduced. [4] **[10]**

5. (a) What is an electrolyte? [2]<<
>>(b) Name the products at the cathode and anode when molten lead(II) bromide is electrolysed. [2]<<
>>(c) Explain why solid lead(II) bromide does not conduct electricity but molten lead(II) bromide does. [3]<<
>>(d) State one industrial use of electrolysis. [3] **[10]**
6. (a) Place potassium, copper and zinc in order of decreasing reactivity. [2]<<
>>(b) Explain why potassium is stored under oil. [2]<<
>>(c) Describe what is observed when magnesium is added to copper(II) sulfate solution. [3]<<
>>(d) State one use of aluminium and a property that makes it suitable. [3] **[10]**
7. (a) Define average speed. [2]<<
>>(b) A learner runs 100 m in 5 s. Calculate the average speed. [3]<<
>>(c) A car travels at constant velocity. What is the resultant force on the car? Explain. [3]<<
>>(d) Name the instrument used to measure time in this experiment. [2] **[10]**
8. (a) Draw the circuit symbols for a cell, a resistor and an ammeter. [3]<<
>>(b) State Ohm's law. [2]<<
>>(c) A resistor of $3\ \Omega$ is connected to a 12 V supply. Calculate the current. [3]<<
>>(d) State one difference between a series and a parallel circuit. [2] **[10]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).